# Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

**Problem B**

**Abstract**

Gravity bends a football's path downward, drag reduces its speed, and Magnus force redirects a spinning flight. We compare gravity, gravity with drag, a constant resultant force, and gravity with aerodynamic forces. The constant-resultant model determines acceleration from three early positions. The aerodynamic model (M4) uses a constant drag coefficient and a linear spin–lift relation to estimate initial velocity, spin direction and aerodynamic parameters.

Three actual penalties supply 39 sampled frames. Three early positions per flight enter inversion, with subsequent measurements used for comparison. For M4, goal-plane errors are 0.360, 0.146 and 0.113 m for the two placement shots and the power shot; the latter has a subsequent root-mean-square path error (RMSE) of 0.093 m. A separate competition-position example quantifies parameter effects: a 1° direction change moves the crossing by about 0.20 m, and the largest grid-search shift under a 2 mm early-position constraint is 0.1052 m. Among 100,000 joint perturbations with fixed mass and radius, 95.772% clear the goal. A same-state curved-shot calculation quantifies rotational damping, while relative airflow describes wind effects.

**Keywords:** Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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# 1 Introduction

## 1.1 Problem Background

A penalty kick is a direct contest between a designated kicker and the opposing goalkeeper. The ball is struck from the penalty mark towards the goal, and the goalkeeper must choose where and when to move. A successful shot combines placement, speed and the ability to conceal its intended direction; a successful save requires an appropriate movement within the short time available.

The importance of this contest has motivated research into both shooting and goalkeeping. Van der Kamp et al. [1] examine how the goalkeeper's anticipation and movement capabilities interact, while Hunter et al. [2] study how kicking speed and technique affect the predictability of shot direction. These questions connect the player's action to the opponent's response and help explain why the fastest shot is not always the most effective choice.

Another line of research concerns the ball after it leaves the foot. Studies examine curved flight and aerodynamic forces[5, 10, 8], the influence of surface structure[11, 9], and the measurement of position, spin and flight trajectories[3, 15, 4]. Together, these studies connect kicking technique, ball motion and the eventual location at which the shot reaches the goal.

## 1.2 Problem Restatement

Problem B concerns the prediction of a penalty trajectory from its early airborne positions. Coordinates are measured in metres, with the penalty mark at the origin, the $x$ axis pointing towards the goal, the $y$ axis lateral and the $z$ axis vertically upward. Time $t$ is measured in seconds. The goal is 11 m from the mark, lies in the plane $x = 11$, and has an opening $-3.66 < y < 3.66$, $0 < z < 2.44$; its width and height are 7.32 m and 2.44 m. At $t = 0$, the ball centre is at $(0, 0, 0.11)$ m.

The task is to develop and test a general method that uses the ball-centre coordinates at $t = 0.05$ s and $t = 0.10$ s to predict its most likely entry point on the goal plane and estimate the uncertainty of that prediction. The specified example has positions $(1.50, 0.45, 0.44)$ m at 0.05 s and $(2.96, 0.86, 0.74)$ m at 0.10 s, and the same method must be applied to these observations.

We examine the airborne motion after the ball has left the kicker's foot. The effect of contact is represented by the initial velocity and spin. The physical forces and governing equations are developed in Section 3; the recovery of initial motion from early positions and the assessment of subsequent predictions are discussed further in Section 4. The specified example, parameter sensitivity and the distribution of goal-plane crossings under stated perturbations are examined in Section 5, connecting the predicted location with its uncertainty. As shown in Figure 1, the analysis links physical modeling, initial-state inference, trajectory prediction and assessment of the predicted crossing.

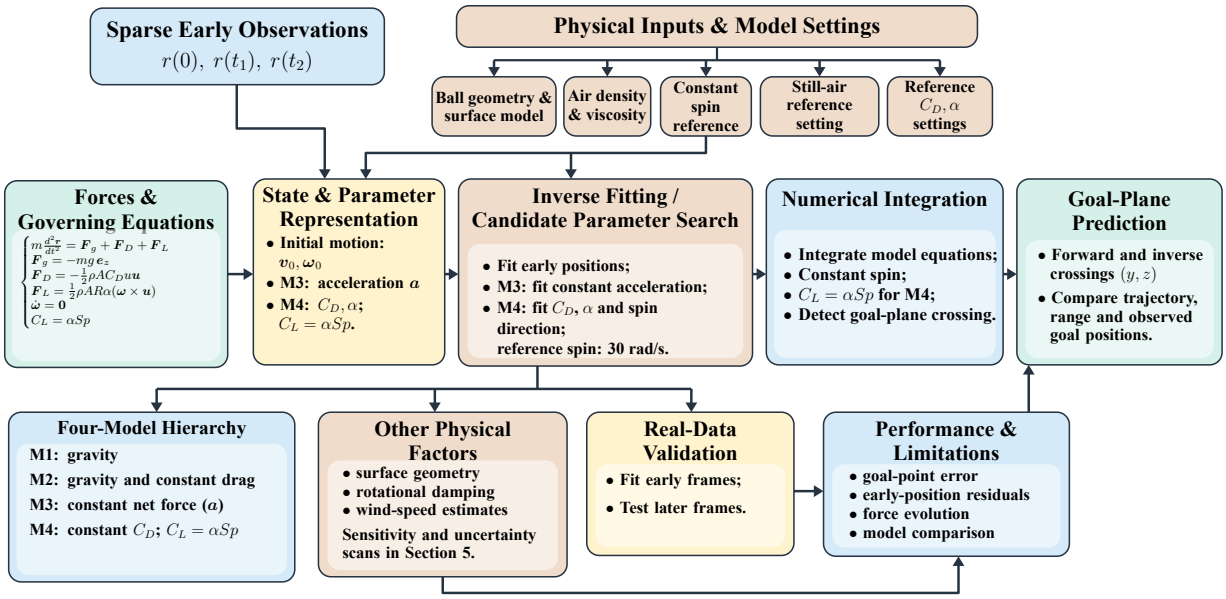



Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

# 2 Preparation of the Models

## 2.1 Assumptions

- **Ball–foot separation.** The airborne motion begins after contact with the kicker has ended. Contact determines the initial velocity and spin, whose role in prediction is discussed further in Section 4.
- **Rigid spherical ball.** The ball retains a spherical shape, with constant mass and radius and coincident mass and geometric centres. The effects of seams and asymmetric surface geometry are discussed further in Section 6.1.
- **Free flight.** Gravity, air resistance and spin-induced lift act while the ball travels through the air. We consider its motion up to the goal plane or the first contact with the ground or another object. The associated geometric conditions are discussed further in Section 3.1.
- **Uniform gravitational field.** The direction and magnitude of gravitational acceleration remain constant over the flight.
- **Uniform air properties.** Air density and viscosity remain uniform during a flight. Their roles in aerodynamic forces are discussed further in Sections 3.2.2 and 3.2.3.
- **Still air.** Air is initially treated as stationary. The effect of wind on relative airflow and force magnitude is discussed further in Section 6.3.
- **Constant spin.** The aerodynamic torque is neglected, so the spin vector remains constant during flight. The influence of rotational damping is discussed further in Section 6.2.

## 2.2 Notations

As shown in Table 1, the position, motion and air properties have distinct roles and units. Each relation below explains the quantities it uses.

Table 1: Principal quantities and competition reference values.

| Symbol | Meaning | Value or unit |
|---|---|---|
| $\boldsymbol{r}, \boldsymbol{v}$ | Ball-centre position and velocity | m; m/s |
| $\boldsymbol{\omega}, \omega$ | Spin vector and its magnitude | rad/s |
| $m, R, A$ | Mass, radius and frontal area | 0.43 kg; 0.11 m; $\pi R^2$ |
| $g$ | Gravitational acceleration | 9.81 m/s$^2$ |
| $\boldsymbol{u}, u$ | Air-relative velocity and its magnitude | m/s |
| $\rho$ | Air density | 1.2 kg/m$^3$ |
| $\nu, \mu = \rho\nu$ | Kinematic and dynamic viscosity | $1.50 \times 10^{-5}$ m$^2$/s; Pa s |
| $Re, Sp$ | Reynolds number and spin parameter | dimensionless |
| $C_D, C_L, \alpha$ | Drag, lift and linear-spin parameters | dimensionless |
| $\lambda_D, \lambda_\alpha$ | Coefficient ratios to reference values | dimensionless |
| $\theta, \boldsymbol{e}_L$ | Spin–airflow angle and Magnus direction | rad; unit vector |
| $q, v$ | Dynamic pressure and ball speed | Pa; m/s |
| $\boldsymbol{Q}, I_b$ | Damping torque and moment of inertia | N m; kg m$^2$ |

We follow Maponi and Piombin [8] for the reference ball parameters and Asai et al. [11] for the reference air density. Kinematic viscosity is fixed at the stated reference value, and dynamic viscosity satisfies $\mu = \rho\nu$. The three match penalties use these same ball and air settings.

# 3 Model Establishment

The four models describe gravitational fall, drag-induced slowing, a constant-force approximation and aerodynamic curvature. The geometry and aerodynamic force relations below lead to the motion equations; the four-model comparison and inverse procedures are developed in Section 4.

## 3.1 Geometry and Goal Plane

As shown in Figure 2, the local horizontal origin is the nominal penalty mark, with $x$ towards the goal, $y$ lateral and $z$ upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m. Section 4.1 converts the measured pitch-centred coordinates.

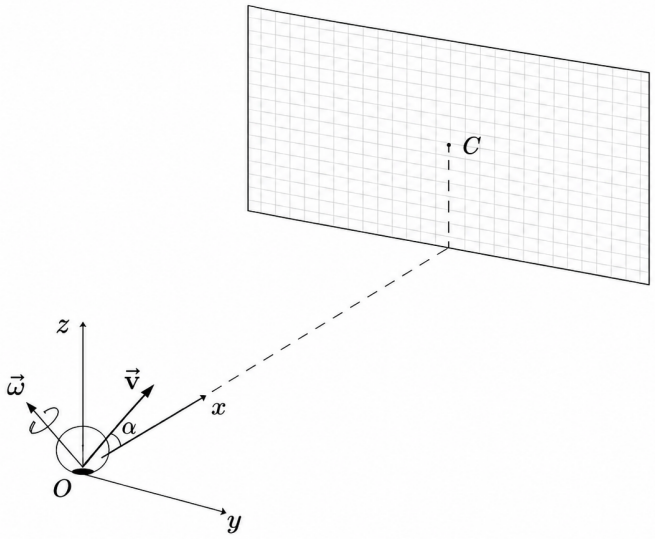



Figure 2: Penalty-kick geometry in local coordinates; $O$ is the nominal penalty mark and $C$ an illustrative goal-plane point.

A clean opening crossing requires $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$. Descending centre height $z = R$ defines ground contact. The coordinates $(y_g, z_g)$ and time $t_g$ denote the first forward crossing of the goal plane. Every model uses these same geometric conditions.

## 3.2 Force Analysis

### 3.2.1 Gravity

Let $\boldsymbol{e}_z$ be the upward vertical unit vector and $\boldsymbol{F}_g$ the gravitational force. For mass $m$ in the uniform field of acceleration magnitude $g$,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

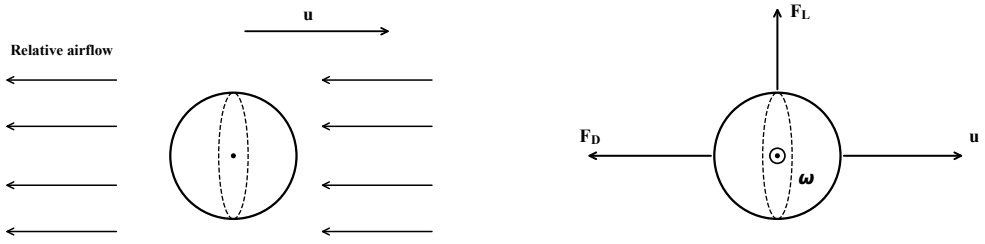
which bends the path downward throughout the flight.

By Archimedes' principle, the displaced air also exerts an upward buoyancy force $\boldsymbol{F}_B$. For ball volume $V_b = \frac{4}{3}\pi R^3$ and air density $\rho$,

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

From these expressions, the combined vertical force is $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. Buoyancy is $1.56\%$ of weight for the reference parameters, so the flight equations retain gravity, drag and Magnus force, with buoyancy treated as a small omitted correction.

To describe the air forces, let $\boldsymbol{v}$ be the ball-centre velocity and $\boldsymbol{v}_{\text{wind}}$ the air velocity, both in m/s. The air-relative velocity is $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$, with speed $u = \|\boldsymbol{u}\|$; the ground-relative speed is $v = \|\boldsymbol{v}\|$. As shown in Figure 3, drag opposes the air-relative motion and spin can redirect the force perpendicular to that motion.




(a) Airflow in the ball frame (b) Example: spin out of the page

Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

### 3.2.2 Aerodynamic Drag

A rapidly moving football displaces air and leaves a wake. The resulting air resistance acts opposite to $\boldsymbol{u}$. Following the aerodynamic force convention used by Goff and Carré [15], we define dynamic pressure $q$ (Pa), frontal area $A$ (m$^2$), and the dimensionless drag coefficient $C_D$. The force magnitude is $F_D = \|\boldsymbol{F}_D\|$. Dynamic pressure sets the force per unit area, while $C_D$ represents the effect of flow and surface structure:

$$q = \tfrac{1}{2}\rho u^2, \qquad A = \pi R^2, \qquad F_D = C_D q A, \qquad \boldsymbol{F}_D = -F_D \frac{\boldsymbol{u}}{u}.$$

Substituting the pressure and area scales into the force direction,

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A C_D u \boldsymbol{u}. \tag{3}$$

According to Asai et al. [11], football drag depends on flow conditions and surface characteristics. We measure the flow regime by the dimensionless Reynolds number $Re = 2Ru/\nu$, where $\nu$ is the kinematic viscosity in m$^2$/s. In the present flight approximation, $C_D$ is constant within each shot; its reference value and estimation are discussed in Section 4.

### 3.2.3 Magnus Lift

A spinning football changes the surrounding airflow and experiences a transverse Magnus force. As shown in Figure 4, rotation makes the relative flow asymmetric. Let $\boldsymbol{\omega}$ be angular velocity, directed along the spin axis by the right-hand rule, and $\omega = \|\boldsymbol{\omega}\|$ its magnitude in rad/s.

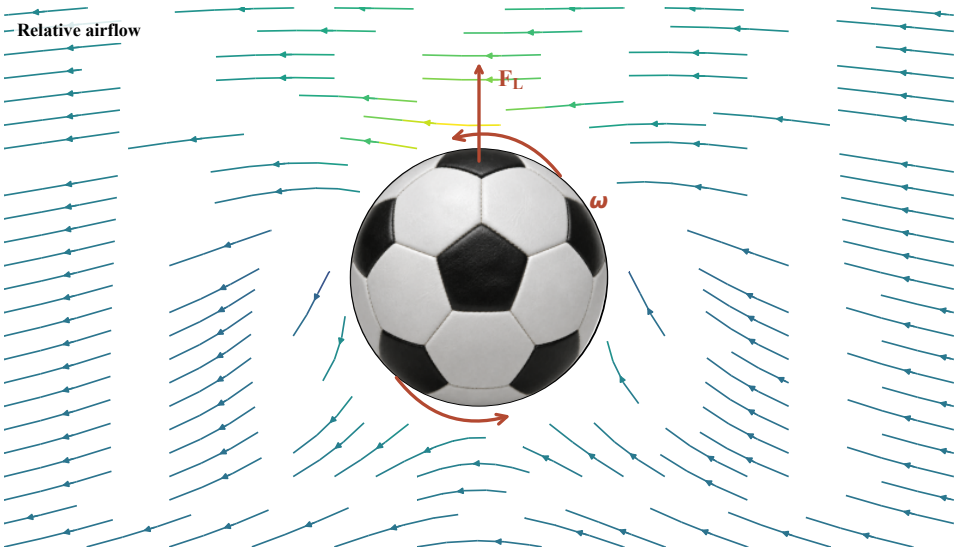



Figure 4: Idealized flow around a spinning football.

Following the lift-coefficient description of Goff and Carré [15], let $C_L$ be the dimensionless transverse lift coefficient. The dimensionless spin parameter $Sp = R\omega/u$ compares the rotating surface speed with the incident airflow speed. We use the linear approximation $C_L = \alpha Sp$, with a dimensionless slope $\alpha$; the reference coefficient is based on Kiratidis and Leinweber [12] and is specified in Section 4. Let $\theta$ be the angle between $\boldsymbol{\omega}$ and $\boldsymbol{u}$, and $\boldsymbol{e}_L$ the unit vector along $\boldsymbol{\omega} \times \boldsymbol{u}$. The lift magnitude is $F_L = \|\boldsymbol{F}_L\|$. Resolving the transverse spin component,

$$F_L = qAC_L \sin\theta, \qquad C_L = \alpha Sp, \qquad Sp = R\omega/u,$$
$$\sin\theta\, \boldsymbol{e}_L = \frac{\boldsymbol{\omega} \times \boldsymbol{u}}{\omega u}, \qquad \boldsymbol{F}_L = F_L \boldsymbol{e}_L.$$

From these relations and $q = \rho u^2/2$,

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho AR\alpha(\boldsymbol{\omega} \times \boldsymbol{u}). \tag{4}$$

The $\sin\theta$ factor extends the transverse-spin relation to a three-dimensional spin direction. The force is zero for zero spin, zero air-relative velocity or parallel spin and velocity. As shown in Figure 5, the force directions distinguish gravitational fall, the small upward buoyancy correction, drag-induced slowing and spin-induced deflection.

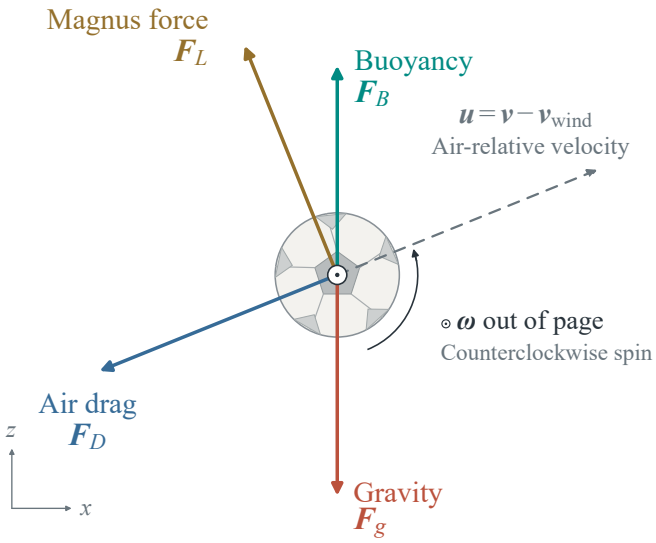



Figure 5: Forces on the football: gravity, buoyancy, drag and Magnus force, with arrows showing their directions for the illustrated state.

## 3.3 Dynamical Equations

Since the football is treated as a rigid body, its motion can be separated into translation of the centre of mass and rotation about that centre. Let $\boldsymbol{L}$ be angular momentum about the centre (kg m$^2$/s), $I_b$ the spherical moment of inertia (kg m$^2$), and $\boldsymbol{Q}$ the resultant external torque (N m). Let $\sum \boldsymbol{F}$ denote the resultant force and an overdot a time derivative. Newton's second law governs translation, while the angular-momentum theorem governs rotation:

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{5}$$

From zero applied torque and constant spherical inertia, $I_b\dot{\boldsymbol{\omega}} = \boldsymbol{0}$, so the spin remains constant. Combining this result with the three retained forces,

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{6}$$

Let $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$ be the initial centre position, velocity and angular velocity, respectively; their subscripts $x, y, z$ denote components in the chosen axes. The initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{7}$$

The initial position fixes the starting location of the flight. The six velocity and spin components describe its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

# 4 Model Solving

Early positions from three actual penalties determine inverse states whose continuations are compared with later observations. Two placement shots and one power shot provide different speeds, curvature directions and flight heights. As shown in Table 2, the four descriptions distinguish gravitational fall, drag-induced slowing, constant acceleration and spin-induced curvature.

Table 2: Four-model hierarchy and inverse quantities.

| Model | Force or motion assumption | Inverse quantities |
|---|---|---|
| M1 | gravity | initial velocity (3) |
| M2 | gravity and drag | velocity and drag coefficient (4) |
| M3 | constant resultant force | velocity and acceleration (6) |
| M4 | gravity, drag and Magnus force | velocity, spin direction, $C_D$, $\alpha$ (7) |

The M3 resultant includes gravity and the other forces together. In M4, $C_D$ and $\alpha$ remain constant during a flight, while $C_L = \alpha Sp$ changes with speed. According to the coefficient relation of Kiratidis and Leinweber [12], we evaluate the reference coefficients at $Re^* = 333793$ and $Sp^* = 0.19$. The superscript $*$ denotes this reference condition: $C_D^* = 0.2415$ and $\alpha^* = C_L(Re^*, Sp^*)/Sp^* = 1.5251$. These values anchor the constant-coefficient approximation.

## 4.1 Inverse Prediction from Three Real Penalties

### 4.1.1 Observations and numerical method

We use the Sportec Open DFL dataset described by Bassek et al. [6] and released by DFL [7]. It combines match events with 25 Hz multi-camera ball tracking. As shown in Table 9, the three selected events comprise two placement shots and one power shot. For each flight, the first selected position after ball–foot separation defines a new time origin; the three inputs occur at $t = 0, 0.04, 0.08$ s. As shown in Table 10, their actual starting positions are retained, and the measured goal crossing supplies the later comparison. The penalty mark is the horizontal origin, with $x$ towards the goal, $y$ lateral and $z$ the centre height. The goal plane is $x = 11$ m. The source records, observed positions and parameter ranges are collected in Appendix A.

Integration uses DOP853 with a maximum step of 0.005 s, relative tolerance $10^{-10}$ and absolute tolerance $10^{-12}$. Forward roots of $x - 11 = 0$ and descending roots of $z - R = 0$ locate the goal crossing and first ground contact.

### 4.1.2 Inversion and reference states

M1 fits velocity, while M2 fits velocity and drag coefficient. For M3, let $\boldsymbol{a}$ be the constant acceleration in m/s$^2$. The three positions satisfy

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \tfrac{1}{2}\boldsymbol{a}t^2. \tag{8}$$

M4 fits three velocity components, two spin-direction angles and two coefficients by minimizing squared coordinate errors at the two later positions. Six deterministic starts are optimized and the

smallest-residual result is selected. As shown in Table 11, the search intervals specify the allowed velocity, spin direction and coefficient ratios. Since $\alpha\boldsymbol{\omega}$ determines rotational deflection, we use a 30 rad/s reference magnitude and fit direction and $\alpha$. Define the unit spin axis $\boldsymbol{n} = \boldsymbol{\omega}/\|\boldsymbol{\omega}\|$, with components $n_x, n_y, n_z$. As shown in Table 12, the fitted speed, coefficients and axis distinguish the initial states of the three shots.

Let $\boldsymbol{r}_i$ be a measured centre position at time $t_i$ and $\widehat{\boldsymbol{r}}(t_i)$ its prediction; $n$ is the number of later positions compared. Hats on the goal coordinates similarly distinguish predictions from observations. The three-dimensional root-mean-square path error $\mathrm{RMSE}_{3D}$ and goal-plane error $E_g$, both in metres, are

$$\begin{aligned} \mathrm{RMSE}_{3D} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n} \|\widehat{\boldsymbol{r}}(t_i) - \boldsymbol{r}_i\|^2}, \\ E_g &= \sqrt{(\widehat{y}_g - y_g)^2 + (\widehat{z}_g - z_g)^2}. \end{aligned} \tag{9}$$

Path comparisons use observations after the fitted times and before first contact or the goal crossing, with sample count $n$ listed in the table. $E_g$ is evaluated when a model crosses the goal plane; its observed counterpart comes from a source frame or interpolation between frames.

## 4.2 Case Analysis and Validation

As shown in Figure 6, the photographs illustrate the two shot styles: Figure 6a shows the placement shot close to the turf, while Figure 6b shows the power shot higher in the air.




**(a)** Illustrative placement shot.




**(b)** Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

As shown in Figure 7, the three spatial paths differ in height and curvature. Open circles are inverse inputs, black points and lines are observations, coloured lines are predictions, and crosses are observed goal crossings.

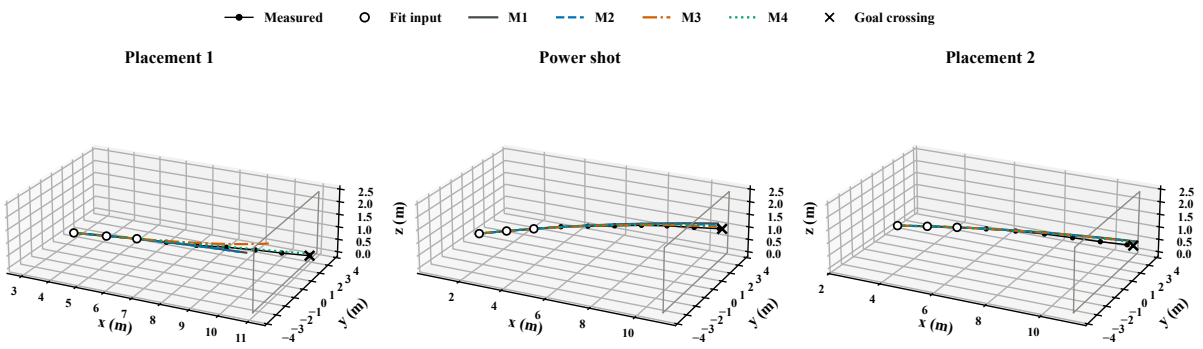



Figure 7: Observed and predicted paths for placement 1, the power shot and placement 2, from left to right. Fitting times are 0, 0.04 and 0.08 s in each case.

### 4.2.1 Case 1: Placement shots

As shown in Figure 8, the four descriptions produce different continuations of the two low flights.

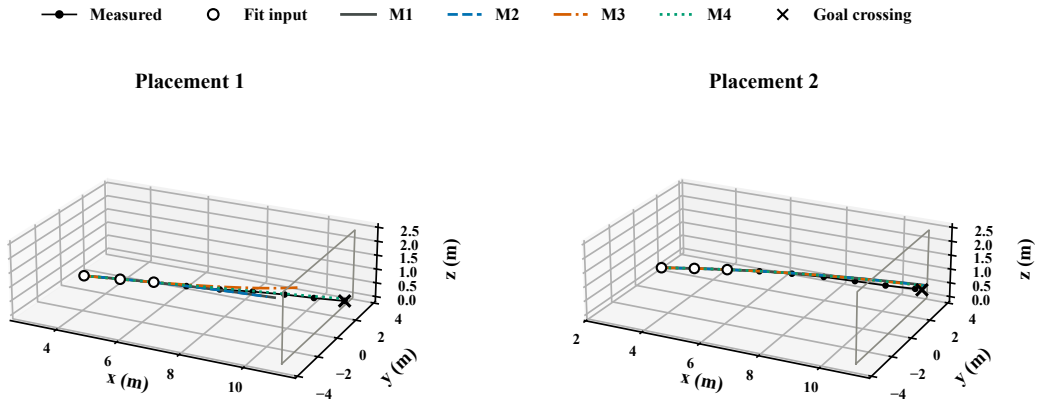



Figure 8: Measured and predicted trajectories for the two placement cases.

Both placement shots travel close to the ground. From the M4 initial states, the starting speeds are 29.20 m/s and 28.34 m/s. For placement 1, M1 and M2 reach the ground before the goal. For M3, the longitudinal acceleration is $a_x = -56.25$ m/s$^2$; from Eq. (8), the maximum forward position is $x = 10.973$ m, short of the goal. M4 reaches the plane with a crossing error of 0.360 m and a subsequent path RMSE of 0.234 m.

All four models reach the goal for placement 2, with crossing errors 0.475, 0.475, 0.155 and 0.146 m. Constant acceleration carries forward the early curvature in this case; M4 reduces the path RMSE from M3's 0.151 m to 0.132 m. Gravity draws a low flight downward, while the vertical Magnus component raises or lowers it according to spin direction. Together they determine whether the ball reaches the goal before contact.

### 4.2.2 Case 2: Power shot

From the M4 initial state, the starting speed is 32.05 m/s. As shown in Figure 9, the spatial path and coordinate histories connect forward deceleration with lateral and vertical curvature.

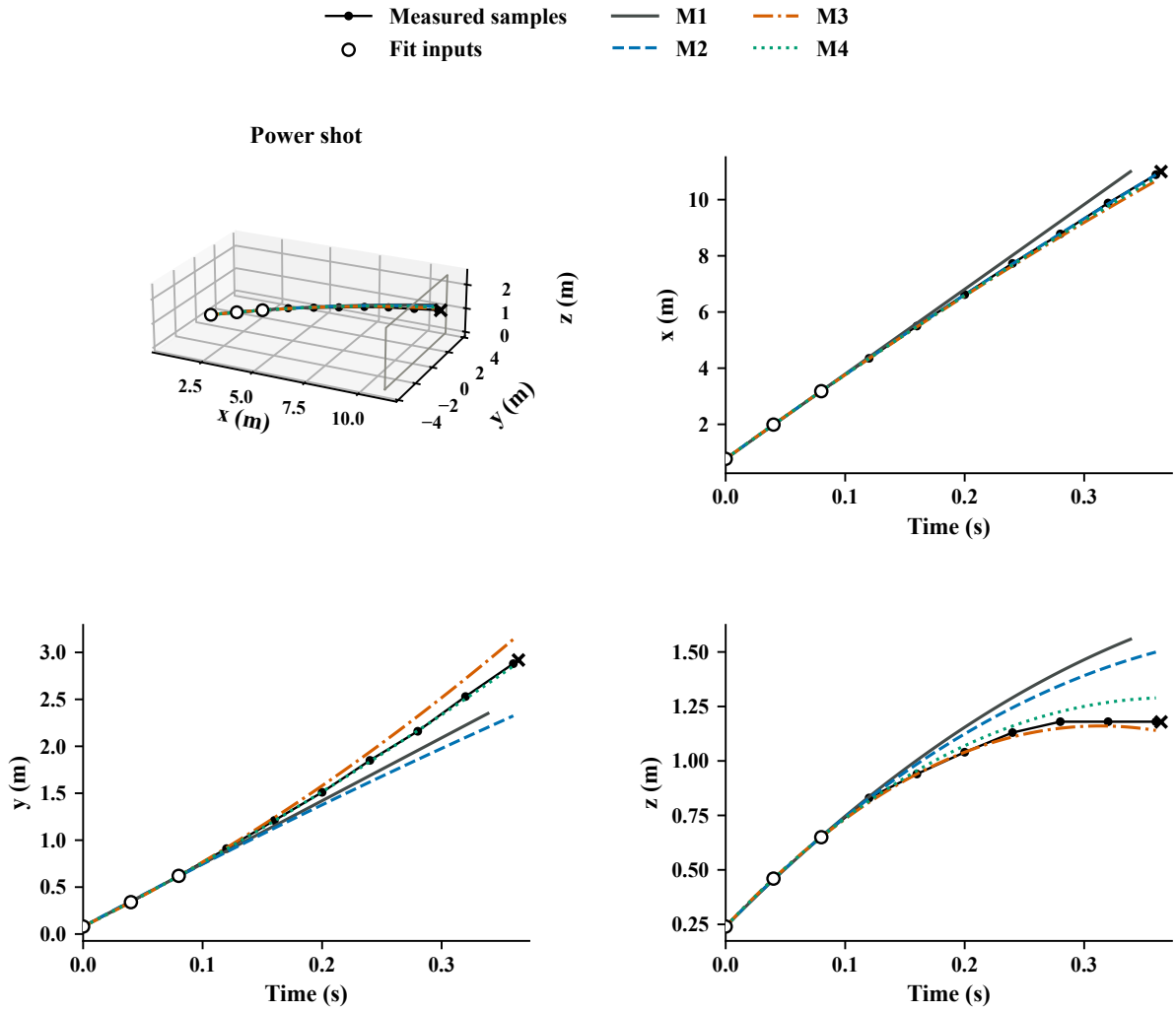



Figure 9: Power-shot path and coordinate histories. Open circles are inverse inputs, black points are observations, and coloured curves are predictions.

M1–M4 have crossing errors of 0.684, 0.657, 0.374 and 0.113 m, and subsequent path RMSE values of 0.412, 0.356, 0.190 and 0.093 m. Lateral motion in M1 and M2 follows the initial velocity; M3 extends constant acceleration. M4 curves the path through changing drag and Magnus forces, giving the smallest values of both errors for this shot.

**Speed and aerodynamic forces.** As shown in Figure 10, the M4 speed and force histories explain how the fitted states develop during flight. Initial drag is 7.58 N for the power shot and 7.05 N and 2.21 N for the placement shots. Drag depends on $C_D v^2$, combining speed with the fitted coefficient,

and decreases as the flight slows. Linear lift depends on $\alpha(\boldsymbol{\omega} \times \boldsymbol{v})$; the angle and direction between spin and velocity determine its lateral and vertical effects.

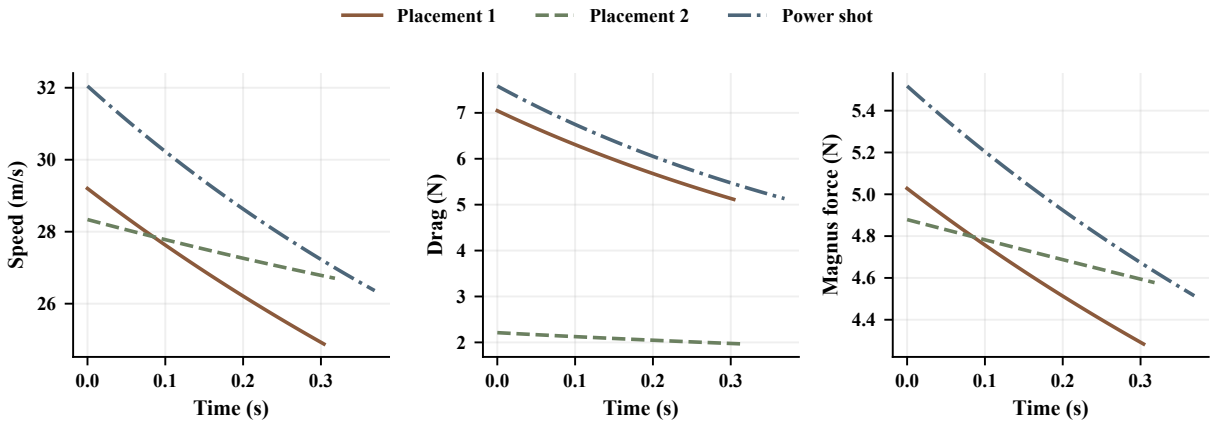



Figure 10: M4 speeds, drag and Magnus force for the three penalties. Curves end at first ground contact or the goal plane.

As shown in Table 3, early fitting residuals, subsequent path errors and goal-plane errors describe different aspects of prediction. The termination status identifies shots that reach the ground or fail to reach the goal plane.

Table 3: Early maximum coordinate residual, subsequent path RMSE/sample count, goal-plane error and termination.

| Event | Model | Residual (mm) | RMSE (m)/$n$ | Goal (m) | Status |
|---|---|---|---|---|---|
| Placement 1 | M1 | 36.00 | 0.244/3 | — | ground |
| Placement 1 | M2 | 26.13 | 0.136/3 | — | ground |
| Placement 1 | M3 | 0.00 | 0.826/6 | — | no crossing |
| Placement 1 | M4 | 25.90 | 0.234/5 | 0.360 | goal |
| Power shot | M1 | 12.00 | 0.412/6 | 0.684 | goal |
| Power shot | M2 | 10.28 | 0.356/7 | 0.657 | goal |
| Power shot | M3 | 0.00 | 0.190/7 | 0.374 | goal |
| Power shot | M4 | 2.90 | 0.093/7 | 0.113 | goal |
| Placement 2 | M1 | 8.00 | 0.300/5 | 0.475 | goal |
| Placement 2 | M2 | 7.01 | 0.320/6 | 0.475 | goal |
| Placement 2 | M3 | 0.00 | 0.151/5 | 0.155 | goal |
| Placement 2 | M4 | 1.12 | 0.132/5 | 0.146 | goal |

# 5 Sensitivity Analysis

The competition's three positions provide a reference example for examining initial motion, aerodynamic coefficients and air conditions. Mass and radius remain fixed at $m = 0.43$ kg and $R = 0.11$ m, with the M4 forces of Section 3. As shown in Table 4, the example uses the two specified early observations together with the initial ball-centre position.

Table 4: Early positions for the sensitivity example.

| $t$ (s) | $x$ (m) | $y$ (m) | $z$ (m) |
|---|---|---|---|
| 0 | 0 | 0 | 0.11 |
| 0.05 | 1.50 | 0.45 | 0.44 |
| 0.10 | 2.96 | 0.86 | 0.74 |

From the inference method in Section 4 and the 30 rad/s reference spin magnitude, the initial motion is

$$\begin{aligned} \boldsymbol{v}_0 &= (30.3784, 9.4074, 6.8974)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.7434, 1.2060, -22.5553)\ \mathrm{rad/s}. \end{aligned} \tag{10}$$

Here $C_D = 0.3623$ and $\alpha = 2.2877$, with a maximum early-coordinate residual of 0.784 mm. The reference flight crosses the goal plane at $(y_g^*, z_g^*) = (2.6159, 1.9526)$ m and $t_g^* = 0.398967$ s.

## 5.1 Effects of individual parameters

One parameter changes at a time. Speed varies by $\pm 5\%$; spin magnitude, $C_D$, $\alpha$ and air density each vary by $\pm 10\%$; velocity and spin-axis azimuth and elevation each vary by $\pm 2^\circ$. Each scan uses 41 equally spaced values, giving 369 flights. Define

$$\begin{aligned} \Delta y_g = y_g - y_g^*, \quad \Delta z_g = z_g - z_g^*, \quad \Delta t_g = t_g - t_g^*, \\ D_g = \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \end{aligned} \tag{11}$$

As shown in Table 5, we compare direction changes of $\pm 1^\circ$ and other changes of $\pm 5\%$, taking the larger $D_g$ and $|\Delta t_g|$ from each positive–negative pair.

Table 5: Goal-plane displacement and arrival-time change under one-factor perturbations.

| Parameter | Perturbation | max $D_g$ (cm) | max $\lvert\Delta t_g\rvert$ (ms) |
|---|---|---|---|
| Launch speed | $\pm 5\%$ | 8.356 | 20.659 |
| Velocity azimuth | $\pm 1^\circ$ | 20.070 | 1.965 |
| Velocity elevation | $\pm 1^\circ$ | 20.218 | 1.692 |
| Spin magnitude | $\pm 5\%$ | 3.873 | 0.287 |
| Spin-axis azimuth | $\pm 1^\circ$ | 1.002 | 0.071 |
| Spin-axis elevation | $\pm 1^\circ$ | 0.852 | 0.084 |
| Drag coefficient | $\pm 5\%$ | 0.620 | 2.274 |
| Lift parameter | $\pm 5\%$ | 3.873 | 0.287 |
| Air density | $\pm 5\%$ | 4.142 | 1.990 |

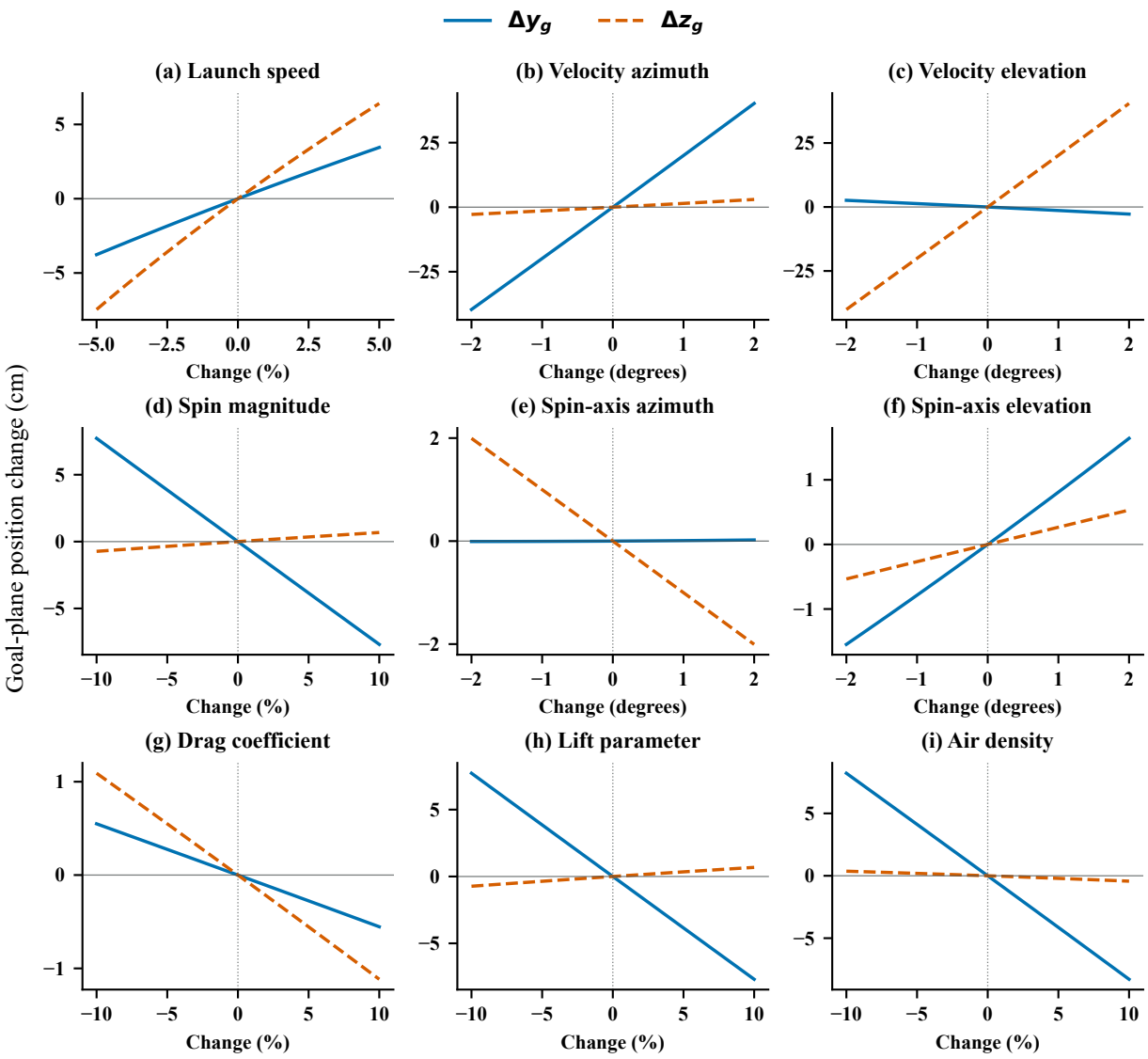



Figure 11: Goal-plane coordinate responses to nine one-factor perturbations. Blue solid curves show $\Delta y_g$ and orange dashed curves $\Delta z_g$, in cm. Vertical scales differ between panels.

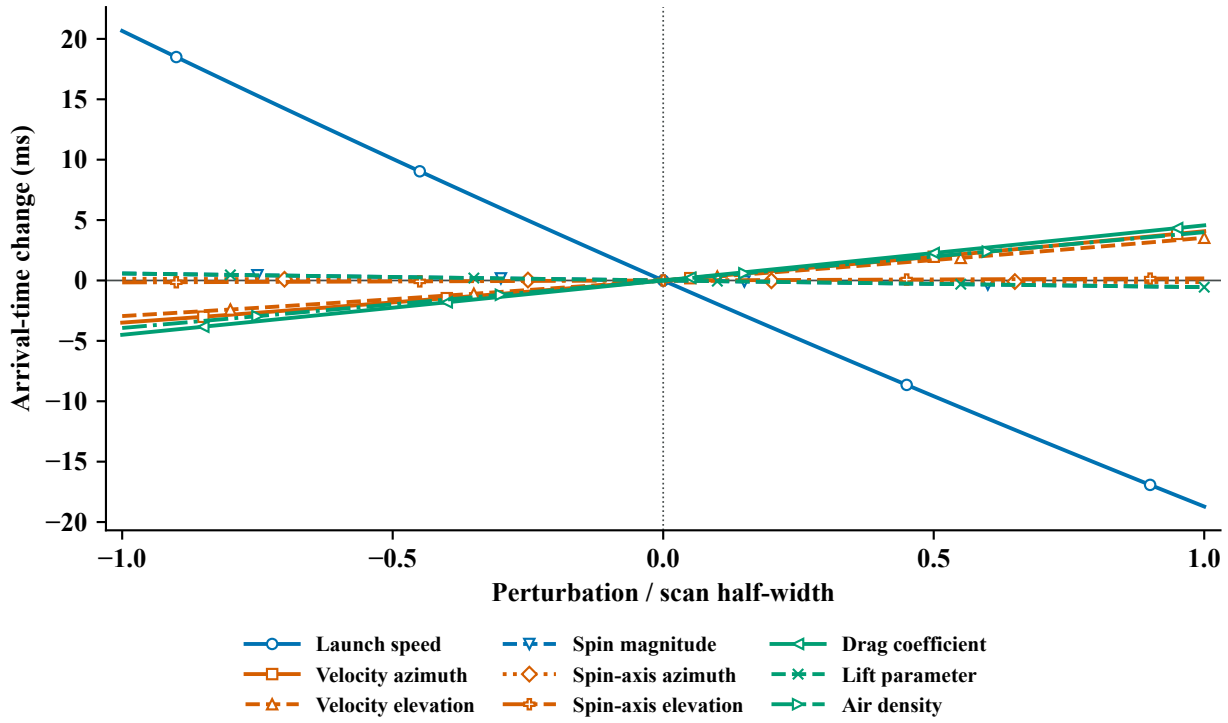



Figure 12: One-factor arrival-time responses. Perturbation divided by scan half-width is plotted horizontally, so $\pm 1$ marks each scan endpoint. The vertical axis is $\Delta t_g$ in ms; positive values mean later arrival.

As shown in Figure 11, a $1^\circ$ change in velocity azimuth or elevation produces a maximum displacement of 20.070 cm or 20.218 cm. As shown in Figure 12, a $5\%$ speed change produces 8.356 cm and 20.659 ms. Speed sets the time available to reach the goal, while direction directly changes lateral and vertical displacement; drag further alters flight time through deceleration.

The linear lift law contains the product of $\alpha$ and spin magnitude. Applying the same relative change to either quantity therefore produces coincident response curves. For any positive dimensionless scale factor $k$,

$$\boldsymbol{\omega}_0' = k\boldsymbol{\omega}_0, \qquad \alpha' = \alpha/k \quad \Longrightarrow \quad \boldsymbol{F}_L' = \boldsymbol{F}_L. \tag{12}$$

This compensation explains the reference spin convention and connects the individual scans with the position-constrained analysis below.

## 5.2 Crossing variation under early-position constraints

As shown in Table 4, the early positions remain fixed while velocity and spin are refitted to examine aerodynamic parameter combinations. With the coefficient references $C_D^*, \alpha^*$, define $\lambda_D = C_D/C_D^*$ and $\lambda_\alpha = \alpha/\alpha^*$. Each multiplier takes 17 values in $[1.2, 1.8]$. At each node, three velocity and three spin components are fitted, using total speed at most 35 m/s, spin components at most 60 rad/s in magnitude, and maximum early-coordinate residual at most 2 mm.

The grid produces 153 compatible flights; 63 lie in the inner square $[1.35, 1.65]^2$. As shown in Table 6, the compatible flights span a range of crossings and arrival times. The largest crossing displacement in the wider scan is 0.1052 m.

Table 6: Ranges of goal-plane crossings compatible with the early positions.

| Range of each multiplier | $y_g$ (m) | $z_g$ (m) | $t_g$ (s) |
|---|---|---|---|
| $[1.35, 1.65]$ | 2.5538–2.6550 | 1.9122–2.0307 | 0.3973–0.4025 |
| $[1.2, 1.8]$ | 2.5478–2.6601 | 1.9075–2.0368 | 0.3973–0.4043 |

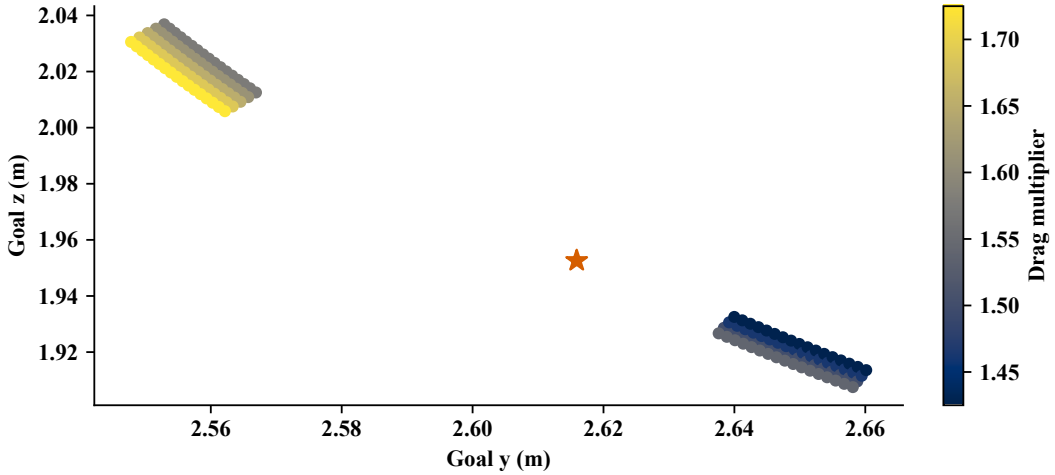



Figure 13: Grid-search crossings under the 2 mm early-position constraint. Colour indicates $\lambda_D$; the orange star is the reference flight.

As shown in Figure 13, different forces and initial states can stay close at early times and separate later, producing the lateral and vertical spread shown here. Refitting the other inverse quantities connects this spread to compensation under the position constraints.

## 5.3 Joint perturbations and Monte Carlo simulation

Nine parameters are sampled independently and uniformly over the full ranges in Section 5.1. Seed 20261108 generates 100000 trajectories. Mass, radius and initial position remain fixed while launch motion and aerodynamic coefficients vary together. Trajectories use fourth-order Runge–Kutta with a 0.002 s step and cubic interpolation of the crossing event. Every trajectory reaches the goal plane before ground contact. As shown in Table 7, the simulated positions and arrival times have both sampled ranges and central percentile intervals.

Table 7: Monte Carlo crossing positions and arrival times (100,000 trajectories).

| Output | Sample minimum–maximum | Central 95% interval, one coordinate |
|---|---|---|
| $y_g$ (m) | 1.9633–3.2143 | 2.1858–3.0505 |
| $z_g$ (m) | 1.4381–2.4646 | 1.5534–2.3535 |
| $t_g$ (s) | 0.3685–0.4346 | 0.3783–0.4218 |

The central 95% intervals are the 2.5th–97.5th percentiles of each output. The mean crossing is $(2.6170, 1.9511)$ m, with lateral and vertical standard deviations 0.2455 m and 0.2360 m. The largest displacement from the reference crossing is 0.801 m.

For a joint region, let $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^T$, with sample mean $\bar{\boldsymbol{q}}$ and covariance $\widehat{\Sigma}$. Let $d_i^2$ be the dimensionless squared distance from the mean and $c_{0.95}$ its empirical 95th percentile. The joint region $\mathcal{E}_{0.95}$ is defined by

$$\begin{aligned} d_i^2 &= (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}), \\ \mathcal{E}_{0.95} &= \{\boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95}\}. \end{aligned} \tag{13}$$

From the empirical cutoff $c_{0.95} = 4.5916$, the ellipse contains 95% of the simulated crossings.

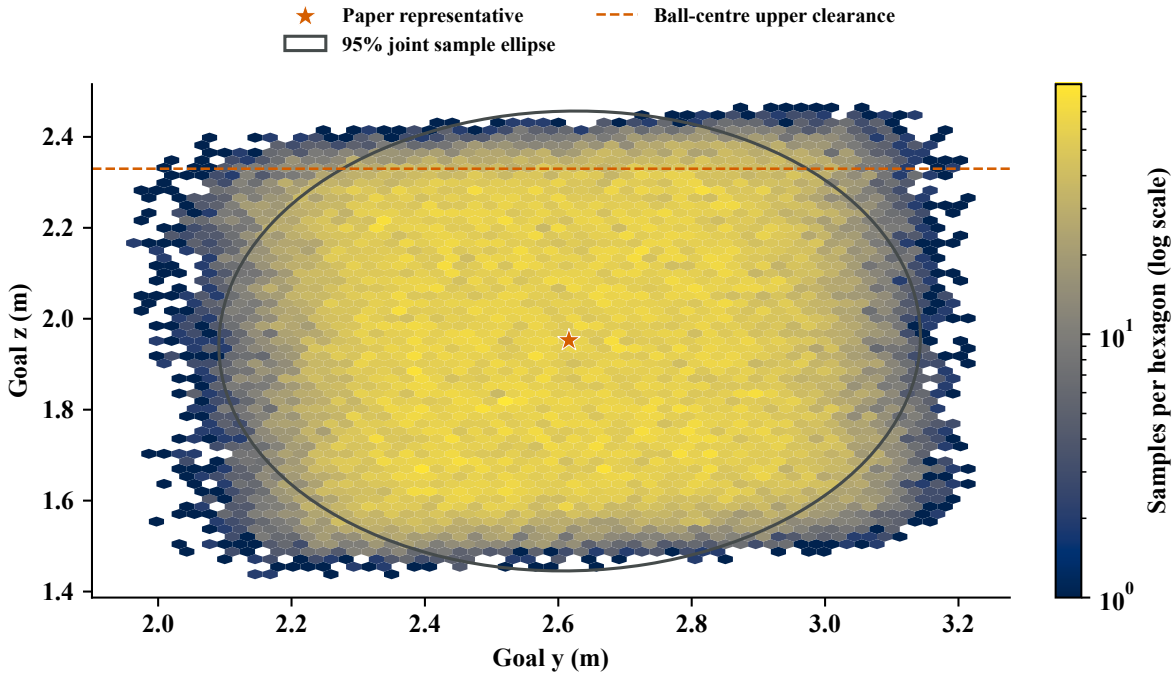



Figure 14: Goal-plane crossings under 100,000 joint perturbations. Hexagon colour indicates sample count, the ellipse indicates the joint 95% region, and the dashed line marks the centre-height clearance limit $z_g = 2.33$ m.

As shown in Figure 14, the joint spread places some crossings above the opening. From the clearance rule of Section 3, 95772 flights (95.772%) pass fully through the opening and 4228 (4.228%) cross too high.

As shown in Figure 15, 400 randomly selected flights connect the input changes with their outcomes. Each line connects nine inputs to $y_g, z_g, t_g$. Inputs use their perturbation limits and outputs the full-sample extrema for linear scaling to 0–1. Blue lines clear the opening, orange lines are too high, and the grey dashed line is the reference state.

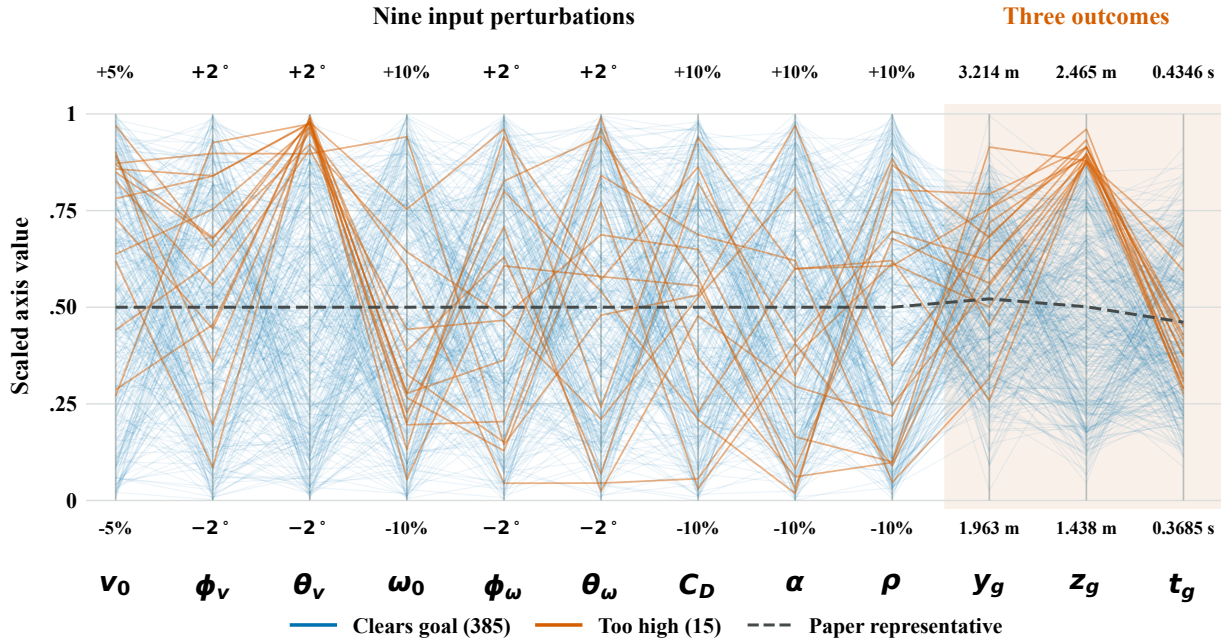



Figure 15: Parallel coordinates linking inputs, goal-plane positions and arrival time. The first nine axes are inputs; three shaded axes are outputs. The displayed subset contains 385 clear flights and 15 high flights.

Following a line reveals the elevation, spin, coefficients and arrival time associated with a high crossing. The spread on the output axes shows lateral and vertical changes together.

As shown in Figure 16, increasing the sample from 50000 to 100000 changes the central-95% interval endpoints by at most 0.264 mm laterally, 0.506 mm vertically and 0.138 ms in time. Two independent seeds each produce 10000 flights. Their largest interval-endpoint differences from the full sample are 5.425 mm laterally, 4.320 mm vertically and 0.328 ms in time.

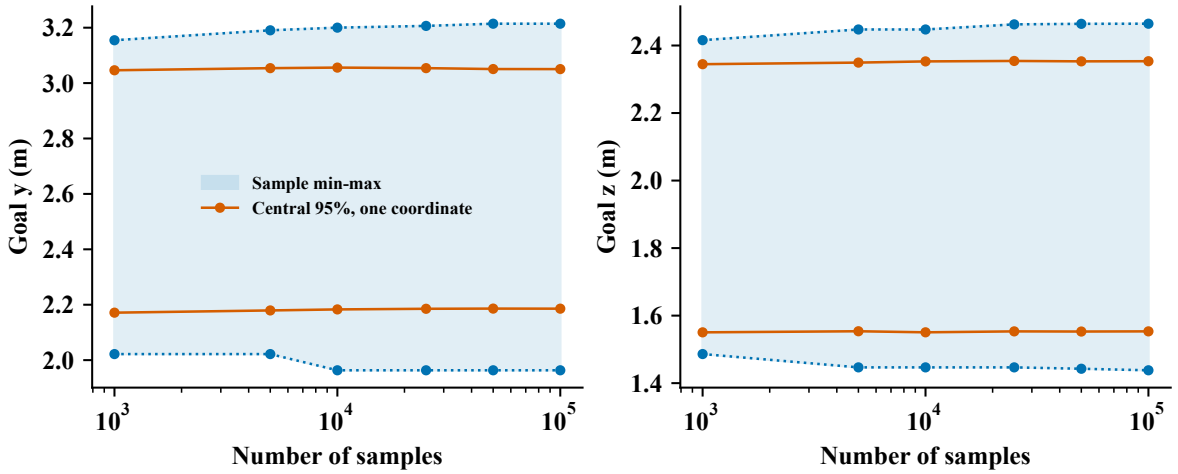



Figure 16: Sample-count effects on crossing ranges and percentile intervals. Blue shading and dotted curves show sampled extrema; orange curves show each coordinate's central 95% interval.

# 6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

## 6.1 Asymmetric Surface Geometry

According to Paudel and Yan [9], full-flight simulations distinguish realistic football geometry from a smooth sphere. As shown in Figure 17, the comparison includes both idealized geometry and actual panel textures; as shown in Figure 17a, we distinguish the seamed and smooth geometries. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



**(a)** Illustrative seamed-football and smooth-sphere comparison, following Paudel and Yan [9].




Fig. 1. Vintage and Adidas Fevernova balls




Fig. 2. Adidas Teamgeist 2006 and 2009 balls




Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

**(b)** Six footballs and surface details, reproduced from Figures 1–3 of Alam et al. [16].

Figure 17: Football geometry, panels and surface textures.

According to Alam et al. [16], wind-tunnel measurements reveal different drag coefficients for six footballs and a smooth sphere over a range of Reynolds numbers. As shown in Figure 17b, these footballs have different panels and textures; as shown in Figure 18, their drag coefficients also differ. The differences illustrate the influence of ball surface structure on aerodynamic drag.

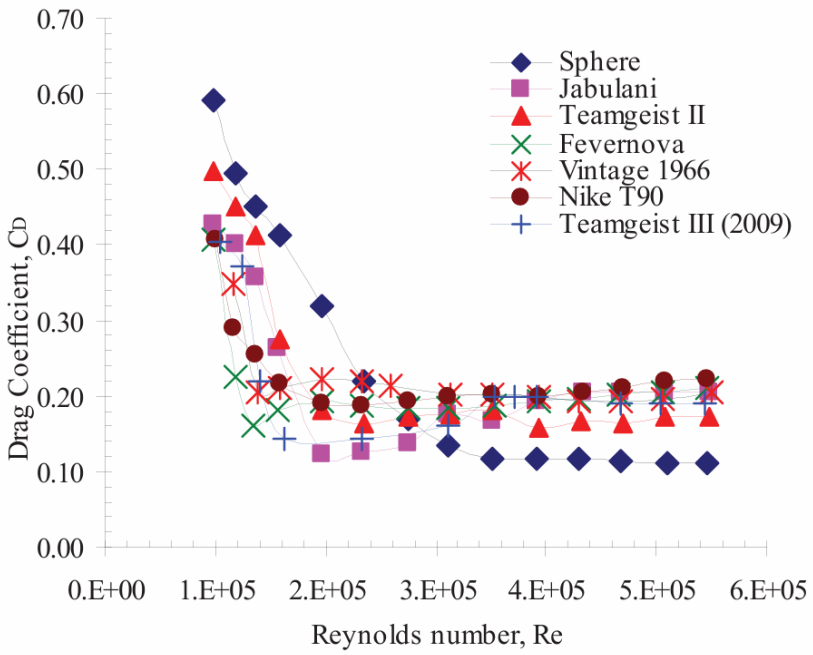



Figure 18: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [16].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through $C_D$ and $C_L$. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

## 6.2 Rotation and Aerodynamic Damping

Aerodynamic torque opposes spin, gradually weakening rotation and changing the Magnus force. Following Grad et al. and the MuJoCo model [13, 14], we use high-Reynolds-number rotational damping: stronger spin produces stronger resistance. The dimensionless angular-drag parameter $C_{\text{angular}} = 1.5$ is the software reference value. Following Maponi and Piombin [8], the inertia is approximated by that of a thin spherical shell. Viscous damping is much weaker than the inertial contribution in this example, so the comparison focuses on the latter.

The reference curved shot has initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with rotational damping. As shown in Figure 19, decreasing spin slightly changes the flight curvature and goal-plane crossing.

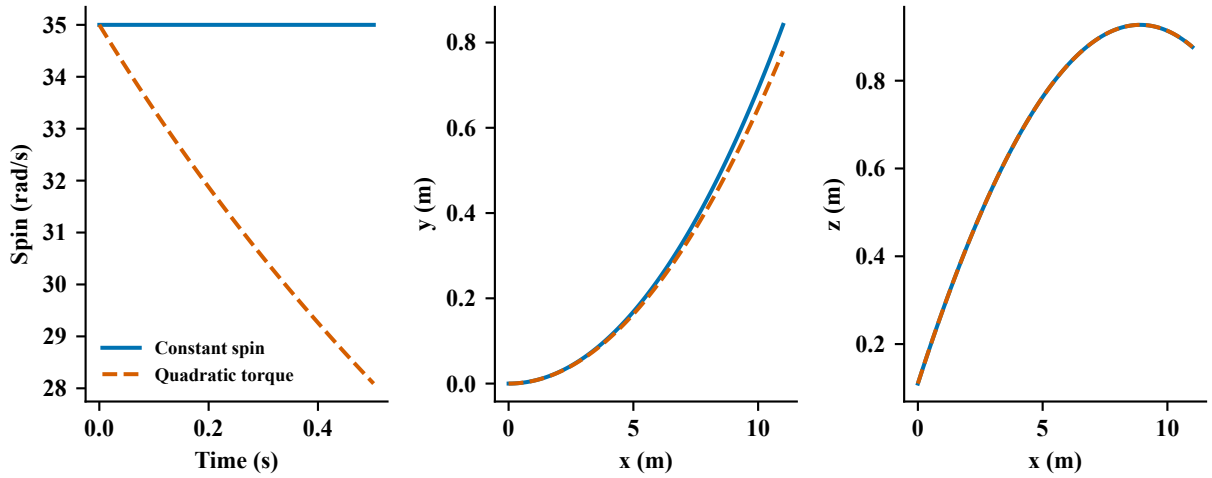



Figure 19: Constant spin and rotational damping under the same initial curved-shot conditions.

Spin falls by 19.740%, while the crossing moves 0.0616 m (0.560% of the 11 m distance) and arrival time changes by -0.066%. Rotational damping therefore has a small endpoint effect in this short reference flight.

## 6.3 Wind Speed and Force Estimates

Following the relative-airflow treatment of Maponi and Piombin [8], wind enters through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$. Let $w$ be wind speed in m/s and $v$ the ball speed. At fixed coefficients, denote the still-air dynamic pressure by $q_0$ and the headwind, tailwind and crosswind values by $q_{\text{head}}$, $q_{\text{tail}}$ and $q_{\text{cross}}$. From the squared relative-speed dependence,

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{14}$$

Crosswind rotates the airflow by $\arctan(w/v)$ and changes the force direction. As shown in Table 8, illustrative winds of 1, 3 and 5 m/s alter dynamic pressure at the 24 m/s curved-shot speed. Let $F_{L,0}$, $F_{L,\text{head}}$ and $F_{L,\text{tail}}$ denote lift magnitudes in still air, headwind and tailwind. From Eq. (4), the M4 linear lift law depends on the first power of relative speed. With fixed spin and $0 \le w < v$,

$$\frac{F_{L,\text{head}}}{F_{L,0}} = 1 + \frac{w}{v}, \qquad \frac{F_{L,\text{tail}}}{F_{L,0}} = 1 - \frac{w}{v}. \tag{15}$$

Table 8: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

| $w$ (m/s) | Headwind (%) | Tailwind (%) | Crosswind (%) | Angle (°) |
|---|---|---|---|---|
| 1 | +8.51 | -8.16 | +0.17 | 2.39 |
| 3 | +26.56 | -23.44 | +1.56 | 7.13 |
| 5 | +46.01 | -37.33 | +4.34 | 11.77 |

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

# 7 Conclusions

The four models describe gravitational fall, deceleration, a constant resultant force and rotational deflection. M3 obtains constant acceleration from three positions. M4 inserts constant $C_D$ and $C_L = \alpha Sp$ into the motion equations, updating drag and Magnus force as speed changes.

For M4, the crossing errors are 0.360, 0.146 and 0.113 m for the three penalties. Its power-shot crossing error and subsequent path RMSE are the smallest among the four models. For placement 1, M3 passes through the early positions but reaches a maximum $x = 10.973$ m, short of the goal; M4 supplies a goal-plane prediction for this low flight.

In the competition-position example, a $1^\circ$ launch-direction change moves the crossing by about 0.20 m, while speed strongly affects arrival time. Spin magnitude and $\alpha$ compensate through the linear lift law. The largest grid-search shift under the 2 mm position constraint is 0.1052 m; 95.772% of 100,000 jointly perturbed flights clear the opening.

Rotational damping shifts the reference curved-shot crossing by 0.0616 m. Wind enters through relative velocity: along the flight direction, drag changes with the squared speed ratio and linear Magnus force with the speed ratio.

# 8 Model Evaluation

## 8.1 Model Strengths

**Physical interpretation.** The model connects the path to three familiar effects: gravity bends it downward, drag reduces speed, and Magnus force redirects a spinning ball. The four-model comparison separates these mechanisms from a constant-acceleration description, making the differences between low placement shots and a higher power shot easier to explain.

**A common framework for inference and prediction.** Three early positions provide the input for estimating initial motion and continuing the flight with the same governing equations. Speed, force histories and goal-plane crossings can therefore be interpreted together. The common coordinate system and termination rules also make comparisons between models and recorded shots consistent.

**Comparison with measurements.** Published aerodynamic measurements support the coefficient choices, while later positions from three actual penalties assess the predicted continuation. The error table, trajectory plots and sensitivity study link the physical assumptions to observable differences. The wind and rotational-damping comparisons further identify when the simpler flight assumptions remain useful.

## 8.2 Model Weaknesses

**Sensitivity of the inferred state.** The fitted velocity, spin direction and aerodynamic parameters depend on the early tracking positions, frame selection and search bounds. Position differences

over short time intervals strongly influence estimated acceleration, especially for flights close to the ground. The reference spin magnitude and the coupling between spin and lift parameter also affect the selected M4 state.

**Simplified aerodynamics and environment.** Constant coefficients during a flight smooth over flow transitions, panel orientation and fluctuating wakes. The spherical, still-air and constant-spin assumptions likewise average effects that may become more important for stronger winds, sustained spin or longer flight. Their relative importance changes with the ball and operating conditions.

**Limited coverage of recorded conditions.** The three penalties cover a modest range of shot types, balls and environments. Broader comparison would benefit from ball-specific aerodynamic data, directly measured spin and more early tracking positions. These additions would strengthen the initial-state estimate and help determine when wind and rotational damping should enter the main prediction.

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# Appendix A: Data Sources, Frame Selection and Parameter Bounds

Following the dataset description by Bassek et al. [6] and the DFL release [7], we retain the source event and frame identifiers. The three selected penalties contain 39 records, with 15, 12 and 12 for the respective events. As shown in Table 9, each case is associated with its player, event identifier and selected samples.

Table 9: Correspondence between case labels and source records

| Case | Event ID | Player | Selected samples |
|---|---|---|---|
| Placement 1 | J03WMX | Dejan Ljubicic | 3, 4, 5 |
| Placement 2 | J03WR9 | Philipp Klement | 3, 4, 5 |
| Power shot | J03WOH | Dawid Igor Kownacki | 2, 3, 4 |

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. Let $(X, Y, Z_{\text{original}})$ denote the published pitch-centred coordinates. The height is interpreted in metres as ball-centre height, and the local coordinates follow from $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

As shown in Table 10, the retained inputs and observed crossings share one local coordinate system. As shown in Table 11, the inverse variables have specified search ranges.

Table 10: Selected early positions and observed goal crossings in local coordinates.

| Event | Source frame | $t$ (s) | $x$ (m) | $y$ (m) | $z$ (m) |
|---|---|---|---|---|---|
| Placement 1 | 152602 | 0.00 | 3.040 | 0.550 | 0.130 |
| Placement 1 | 152603 | 0.04 | 4.190 | 0.820 | 0.150 |
| Placement 1 | 152604 | 0.08 | 5.250 | 1.070 | 0.180 |
| Placement 1 goal | 152610 | 0.3200 | 11.0000 | 2.7500 | 0.1600 |
| Placement 2 | 176245 | 0.00 | 2.570 | 0.850 | 0.340 |
| Placement 2 | 176246 | 0.04 | 3.640 | 1.200 | 0.410 |
| Placement 2 | 176247 | 0.08 | 4.710 | 1.530 | 0.470 |
| Placement 2 goal | 176253–176254 | 0.3295 | 11.0000 | 3.0697 | 0.4657 |
| Power shot | 128955 | 0.00 | 0.770 | 0.080 | 0.240 |
| Power shot | 128956 | 0.04 | 1.990 | 0.340 | 0.460 |
| Power shot | 128957 | 0.08 | 3.180 | 0.620 | 0.650 |
| Power shot goal | 128964–128965 | 0.3643 | 11.0000 | 2.9188 | 1.1789 |

Table 11: Component bounds and force-correction settings for inversion.

| Quantity | Search interval |
|---|---|
| $v_{x0}$ (M1, M2, M4) | $[0, 35]$ m/s |
| $v_{y0}, v_{z0}$ (M1, M2, M4) | $[-35, 35]$ m/s |
| $\|\boldsymbol{\omega}_0\|$ (M4) | 30 rad/s |
| Azimuth, elevation (M4) | $[-\pi, \pi], [-\pi/2, \pi/2]$ rad |
| $C_D/C_D^*, \alpha/\alpha^*$ | $[0.5, 1.5]$ |

As shown in Table 12, the M4 states differ in speed, coefficients and unit spin axis $\boldsymbol{n} = \boldsymbol{\omega}/\|\boldsymbol{\omega}\|$.

Table 12: Fitted M4 states at the selected starting frames.

| Event | $v_0$ (m/s) | $C_D$ | $\alpha$ | $n_x$ | $n_y$ | $n_z$ |
|---|---|---|---|---|---|---|
| J03WMX | 29.20 | 0.3623 | 2.2877 | 0.2280 | -0.9728 | 0.0413 |
| J03WR9 | 28.34 | 0.1208 | 2.2877 | 0.1439 | -0.2711 | -0.9517 |
| J03WOH | 32.05 | 0.3238 | 2.2877 | -0.2276 | 0.3170 | 0.9207 |

# Reproduction Code

The following excerpts evaluate the force components and state derivatives used in Eq. (6). Initial-state inference and subsequent trajectory prediction use these same physical relations and coefficient settings.

```
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```